Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester Theory B. Tech (All branches) Examination

April-May 2018

ME 102 Fundamentals of Mechanical Engineering

Date: 07.05.2018, Monday **Time: 01:30 p.m. To 04.30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 (a) What is prime mover? Give any example of prime mover. **[02]**

Q – 1 (b) Name any two parts of single acting reciprocating pump along with its uses. **[02]**

Q – 1 (c) Give advantages and disadvantages of gaseous fuels. **[03]**

Q – 2 **Answer any two.** **[14]**

(a) Classify the Internal Combustion engine.

(b) A Bajaj auto rickshaw has two stroke internal combustion engine having stroke length of 140 mm and cylinder diameter 90 mm. Its mean effective pressure is 5.4 bar and speed of the engine is 1000 rpm. Determine brake power of the engine. Assume mechanical efficiency as 85%.

(c) Discuss why petrol engines are called as S.I. engines? And why diesel engines are called as C.I. engine?

(d) Explain the working of brakes which are used in a bicycle with neat sketch.

Q – 3 **Answer any two.** **[14]**

(a) Explain uses of compressed air.

(b) Explain working of gear pump with its neat sketch.

(c) Give minimum four applications of refrigeration. Write down three properties of a good refrigerant.

(d) With the help of neat sketch explain the working of vapour compression refrigeration cycle.

SECTION – II

Q – 4 (a) Explain Boyle's law with p-V diagram. **[02]**

Q – 4 (b) If any steam has dryness fraction of 0.8, then what can you say about its type? What amount of water particle will be there in 1 kg of this steam? **[02]**

Q – 4 (c) What are the types of belt? **[03]**

Q – 5 **Answer any two.** **[14]**

(a) Draw the figure of Babcock and Wilcox boiler and explain its working.

(b) Compare water tube boiler and fire tube boiler.

(c) How the air preheater works? Draw its neat sketch.

(d) Enlist advantages and disadvantages of gear drives.

Q – 6 **Answer any two.** **[14]**

(a) Prove pV=mRT.

(b) 0.5 m^3 of air at pressure 1 bar is compressed at constant temperature until its volume is 0.2 m^3. What is then its pressure? The air is then heated under constant pressure process until its temperature 75° C. Determine its volume and mass. Assume compression take place at 20° C. and R = 287 J / kg K.

(c) Draw p-V diagram for steam formation at constant pressure. Show different terms on it.

(d) Prove that dryness fraction $X = X_1X_2$ for combined calorimeter.

***